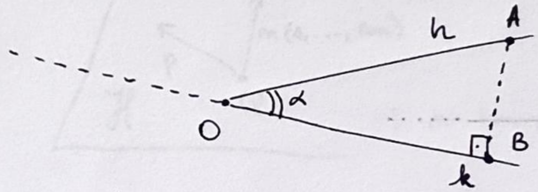
Lecture 3 – Geometry.

10.03.2026

Scalar. Product. Angles

Def: Let h, k be two rays emmanating from a point O.



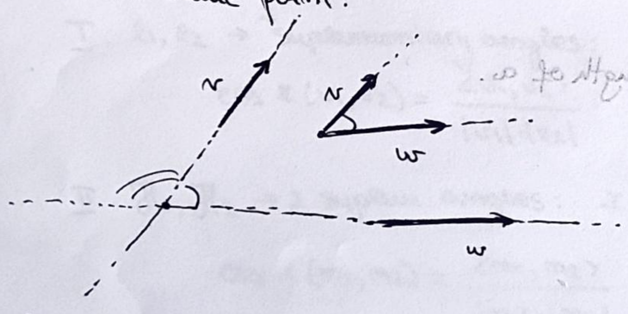
two rays $\Rightarrow$ angle.

$\Rightarrow$ it defimes $\measuredangle(h,k)$.

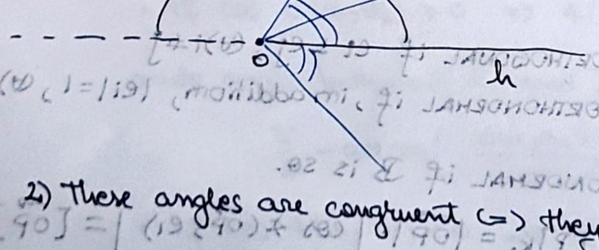
$\sin\alpha = \dfrac{|AB|}{|OA|}$ $\quad \dfrac{(\text{cat. op.})}{(\text{ip})}$.

$$\cos\alpha = \begin{cases} 0, & \text{angle is right angle} \\ \dfrac{|OB|}{|OA|}, & \text{angle is acute} \\ -\dfrac{|OB|}{|OA|}, & \text{angle is obtuse.} \end{cases} \quad \left(\frac{\text{cat. al}}{\text{ip}}\right)$$

Def: Let $v, w \in \mathbb{V}$ (two vectors). The angle $\measuredangle(v,w)$ is any angle given by representatives in the same point.

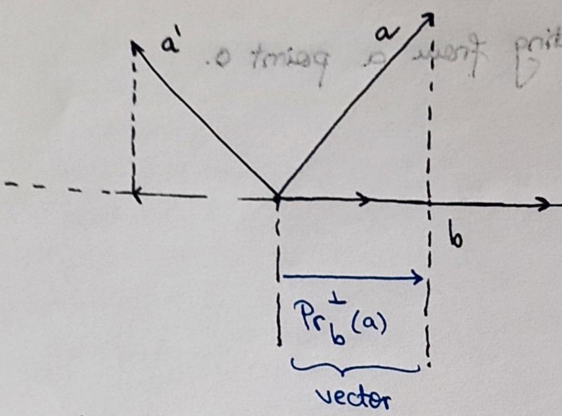


Obs! 1) these angles are in bijection with the semicircle.



2) These angles are congruent $\Leftrightarrow$ they have the same cosine.

Orthogonal projection of vectors.



$$\underbrace{Pr_b^{\perp}(a)}_{\text{vector}} = \underbrace{pr_b^{\perp}(a)}_{\text{scalar}} \cdot \underbrace{\frac{b}{|b|}}_{\text{unit vector}}.$$

$Pr_b^{\perp} : V \to V$

$pr_b^{\perp} : V \to \mathbb{R}$ (length)

} orthogonal projection maps.

Prop. 3.4.
These projection maps are linear and, moreover,

$$\boxed{\cos \measuredangle(a,b) = \frac{pr_b^{\perp}(a)}{|a|} = \frac{pr_a^{\perp}(b)}{|b|}}$$

$|a| \leftarrow$ length of a.

Orthonormal frames.

$\mathbb{R}^n$.

$K(O, B)$ — O: origin, B: basis.

A basis $B = (e_1, \dots, e_n)$ is
- ORTHOGONAL if $e_i \perp e_j$; $(\forall) i \neq j$
- ORTHONORMAL if, in addition, $|e_i| = 1$, $(\forall) i$.

A frame is ORTHOGONAL / ORTHONORMAL if B is so.

In an ORTHONORMAL frame, $[P]_K = |\vec{OP}| \cdot \begin{vmatrix} \cos \measuredangle(\vec{OP}, e_1) \\ \dots \\ \cos \measuredangle(\vec{OP}, e_n) \end{vmatrix} = [\vec{OP}]_B$.

$P(x_1, \dots, x_n)$ point.

Oriented angles.

↳ only in the plane (not space).

$$\measuredangle_{or}(h,k) = \begin{cases} \measuredangle(h,k), & \text{if } (\vec{OA}, \vec{OB}) \text{ is right-oriented.} \\ -\measuredangle(h,k), & \text{otherwise.} \end{cases}$$

$\exists$ a bijection between the set of oriented angles and the unit circle.

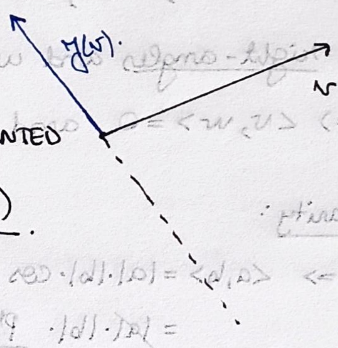
J-operator

$J : V^2 \to V^2$. defined by $J(v)$ is the unique vector s.t.:

$J(v) \perp v$.

$|J(v)| = |v|$.

$(v, J(v))$ right-oriented.



Def: $a, b \in V^2$.

$$\sin \measuredangle_{or}(a,b) = \frac{pr^{\perp}_{J(a)}(b)}{|b|}.$$ ORIENTED

$$\cos \measuredangle_{(or)}(a,b) = \frac{pr^{\perp}_{a}(b)}{|b|}$$

ORIENTED / UNORIENTED.

Scalar product.

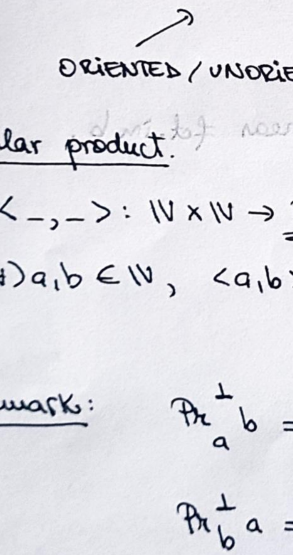
$\langle \_, \_ \rangle : V \times V \to \mathbb{R}$.

$$(\forall) a,b \in V, \quad \langle a,b \rangle = \begin{cases} 0, & \text{if } a = 0 \text{ or } b = 0. \\ |a| \cdot |b| \cdot \cos \measuredangle(a,b), & \text{otherwise.} \end{cases}$$

Remark:

$$Pr^{\perp}_{a} b = \frac{\langle a,b \rangle}{\langle a,a \rangle} \cdot a$$

$$Pr^{\perp}_{b} a = \frac{\langle b,a \rangle}{\langle b,b \rangle} \cdot b$$



$$Pr^{\perp}_{a}(b) = pr^{\perp}_{a}(b) \cdot \frac{a}{|a|}$$

$$= |b| \cdot \cos \measuredangle(a,b) \cdot \frac{a}{|a|} = \frac{|b| \cdot |a| \cdot \cos \measuredangle(a,b)}{|a|^2} \cdot a$$

$$= \underbrace{\frac{pr^{\perp}_{a}(b)}{|b|}}_{\cos} \cdot |b| \cdot \frac{a}{|a|}$$

Properties of scalar prod:

1) Billinear:

$(\forall)\, a, b \in \mathbb{R}$

$(\forall)\, v, w, u \in \mathbb{V}^2$: $\langle av + bw, u\rangle = a\langle v, u\rangle + b\langle w, u\rangle$

$\langle v, aw + bu\rangle = a\langle v, w\rangle + b\langle v, u\rangle$

2) Symmetric:

$(\forall)\, v, w \in \mathbb{V}^2$: $\langle v, w\rangle = \langle w, v\rangle$.

3) Positive definite:

$(\forall)\, v \in \mathbb{V}^2,\ v \neq 0 \Rightarrow \langle v, v\rangle > 0$.

4) Recognizes right-angles and unit lengths.

$v \perp w \Leftrightarrow \langle v, w\rangle = 0$ and $|v| = 1 \Leftrightarrow \langle v, v\rangle = 1$.

Prove bilinearity:

by symm. $\Rightarrow$ $\langle a, b\rangle = |a| \cdot |b| \cdot \cos \sphericalangle(a, b)$

$= \cancel{|a|} \cdot |b| \cdot \dfrac{pr_b^{\perp}(a)}{\cancel{|a|}}$

$= \underbrace{|b|}_{ct.} \cdot \underbrace{pr_b^{\perp}(a)}_{\text{linear fct. in } b}$ $\Rightarrow$ linear fct. in b.

?

Prop. 3.11: $K = (O, \mathcal{B})$ orthonormal frame.

$a(a_1, \ldots, a_m)$

$b(b_1, \ldots, b_m)$

$\langle a, b\rangle = a_1 b_1 + \ldots + a_m b_m$.

$|a| = \sqrt{a_1^2 + a_2^2 + \ldots + a_m^2}$.

$\cos \sphericalangle(a, b) = \dfrac{\langle a, b\rangle}{|a| \cdot |b|} = \dfrac{a_1 b_1 + \ldots + a_m b_m}{\sqrt{a_1^2 + \ldots + a_m^2} \cdot \sqrt{b_1^2 + \ldots + b_m^2}}$

$a \perp b \Leftrightarrow a_1 b_1 + \ldots + a_m b_m = 0$.

Proof: $\boxed{\langle a, b\rangle = a_1 b_1 + \ldots + a_m b_m}$

$\langle a, b\rangle = \langle a_1 e_1 + a', b_1 e_1 + b'\rangle = a_1 \langle e_1, b_1 e_1 + b'\rangle + \cancel{b_1}\langle a', b_1 e_1 + b'\rangle$.

The orthonormal frame is $K = (O, \mathcal{B} = (e_1, \ldots, e_m))$.

$= a_1 b_1 \cdot \overset{1}{\langle e_1, e_1\rangle} + a_1 \cdot \overset{0}{\langle e_1, b'\rangle} +$

$+ b_1 \cdot \langle a', e_1\rangle + \langle a', b'\rangle$.

$= a_1 b_1 + \langle a', b'\rangle \Rightarrow$ induction.

so

$a = a_1 e_1 + \underbrace{a_2 e_2 + \ldots + a_m e_m}_{a'}$

$b = b_1 e_1 + b'$.

Gram - Schmidt algorithm.

$K=(O,\mathcal{B})$. non-orthonormal frame. $\longrightarrow$ $K'(O,\mathcal{B}')$ orthonormal frame.

$\mathcal{B}=(\underbrace{e_1,e_2}_{\perp ?},\dots,e_n)$ basis of $\mathbb{V}^n$. $\mathcal{B}'$ => orthonormal basis. "$(\underbrace{e_1',e_2'}_{\perp ✓},\dots,e_n')$".

• Projection of e_i onto e_j : $\dfrac{\langle e_j,e_i\rangle}{\langle e_j,e_j\rangle}\cdot e_j$.

1) Construct an orthogonal basis $e_1',e_2',\dots,e_n'$:

$$e_1' = e_1 = e_1$$

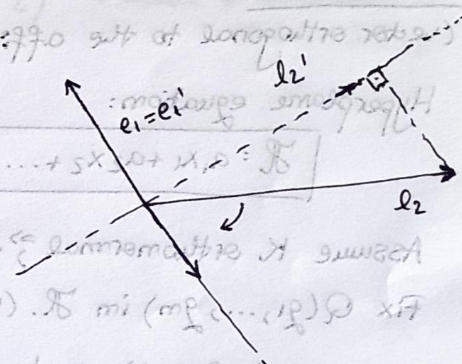
$$e_2' = e_2 - \frac{\langle e_1',e_2\rangle}{\langle e_1',e_1'\rangle}\cdot e_1' = e_2 - Pr^{\perp}_{e_1'}(e_2)$$

$$e_3' = e_3 - \frac{\langle e_1',e_3\rangle}{\langle e_1',e_1'\rangle}\cdot e_1' - \frac{\langle e_2',e_3\rangle}{\langle e_2',e_2'\rangle}\cdot e_2' = e_3 - Pr^{\perp}_{e_1'}(e_3) - Pr^{\perp}_{e_2'}(e_3).$$

$\vdots$

2) Normalize the vectors to obtain the basis:

$$\mathcal{B}' = \left\{\frac{e_1'}{|e_1'|},\dots,\frac{e_n'}{|e_n'|}\right\}.$$



∞ => "process"

finite => "algorithm".

Proof:

Pr.3.13: The basis $\mathcal{B}'$ obtained from the basis $\mathcal{B}$ with the Gram-Schmidt alg. is an orthonormal basis.

$\forall K$. → Consider $\mathbb{V}_K$ generated by $B_K=(e_1,\dots,e_K)$.

=> We show that $B_K'=(e_1',\dots,e_K')$ is an orthonormal basis for $\mathbb{V}_K$.

• if $K=1$ => nothing to prove.

• let $K>1$. consider:

$$e'_{K+1} = e_{K+1} - \underbrace{\sum_{j=1}^{K} Pr_{e_j'}(e_{K+1})}_{v}$$

• notice that $e_{K+1}\neq v$, since e_{K+1} is lin. ind. of $\mathbb{V}_K$.

Linear combination of the first K vectors: $\dfrac{\langle e_j',e_{K+1}\rangle}{\langle e_j',e_j'\rangle}\cdot e_j'$.

$(e_1', \dots, e_k', e'_{k+1})$ is indeed a basis of $\mathbb{V}_k$.

? $e'_{k+1} \perp e_i$, $i \le k$.

$$\langle e'_{k+1}, e_i' \rangle = \left\langle e_{k+1} - \sum_{j=1}^{k} \frac{\langle e_j', e_{k+1}\rangle}{\langle e_j', e_j'\rangle} \cdot e_j', e_i' \right\rangle$$

$$= \langle e_{k+1}, e_i'\rangle - \sum_{j=1}^{k} \frac{\langle e_i', e_{k+1}\rangle}{\langle e_j', e_j'\rangle} \cdot \underbrace{\langle e_j', e_i'\rangle}_{\substack{1 \text{ if } i=j \\ 0 \text{ otherwise.}}}$$

$$= \langle e_{k+1}, e_i'\rangle - \frac{\langle e_i', e_{k+1}\rangle}{\cancel{\langle e_i', e_i'\rangle}} \cdot \cancel{\langle e_i', e_i'\rangle}^{1}$$

$= 0.$

(because scalar product is symmetric).

Normal vectors

(vector orthogonal to the affine space).

Hyperplane equation:

$$\boxed{\mathcal{H} : a_1x_1 + a_2x_2 + \dots + a_mx_m = b.} \quad (1)$$

Assume K orthonormal $\Rightarrow \mathcal{B} = (e_1, \dots, e_m)$ orthonormal

Fix $Q(q_1, \dots, q_m)$ in $\mathcal{H}$. (1)

$\Rightarrow Q$ satisfies (1) $\Rightarrow b = a_1q_1 + a_2q_2 + \dots + a_mq_m$.

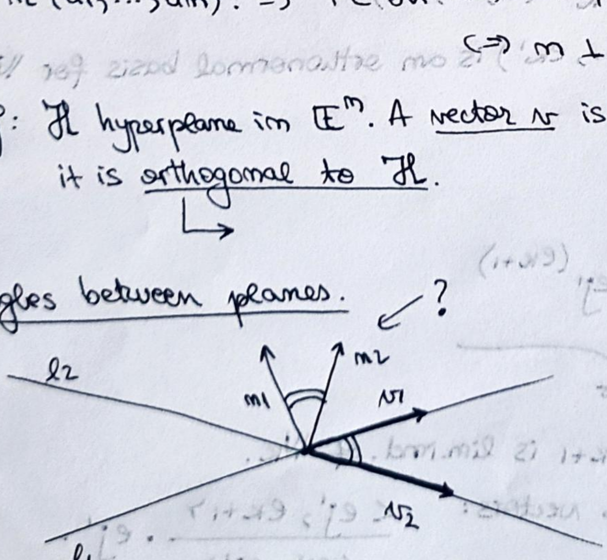
$(\forall) P \in \mathcal{H}$: $a_1(p_1 - q_1) + a_2(p_2 - q_2) + \dots + a_m(p_m - q_m) = 0$.

$n(a_1, \dots, a_m)$. $\Rightarrow P \in \mathcal{H} \Leftrightarrow \langle n, \overrightarrow{QP}\rangle = 0$

$\Leftrightarrow n \perp \overrightarrow{QP}$.

Def: $\mathcal{H}$ hyperplane in $\mathbb{E}^m$. A vector v is called a normal vector of $\mathcal{H}$ if it is orthogonal to $\mathcal{H}$.

Angles between planes. ?



$l_1, l_2 \to$ lines in $\mathbb{E}^2$.

$v_1 \to$ direction vector for l_1.

$v_2 \to$ direction vector for l_2.

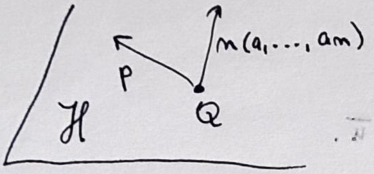
angles described by l_1, l_2:

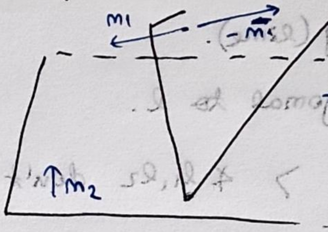
$$\underbrace{\measuredangle(v_1, v_2),\ \measuredangle(-v_1, v_2)}_{\text{suplementary.}}$$

$$\cos \measuredangle(v_1,v_2) = \frac{\langle v_1, v_2\rangle}{|v_1|\cdot|v_2|} \to \cos \measuredangle\, n_1, n_2 \qquad \frac{\langle n_1,n_2\rangle}{|n_1|\cdot|n_2|} \; ?$$

↖ normal vectors.

2 options for calculating one of the angles between 2 lines ?





→ one angle between 2 planes is the angle between two normal vectors.

• I know v_1 for one line and n_2 for the other line, then :
(direction vector) (normal vector)

$$\frac{\pi}{2} - \arccos\left(\left|\frac{\langle v_1, v_2\rangle}{|v_1|\cdot|v_2|}\right|\right) \in \left[0, \frac{\pi}{2}\right) \to \text{acute } \measuredangle \text{ between those lines.}$$

↳ Generalization:

$l_1, l_2 \to$ lines.

$v_1, v_2 \to$ direction vectors.

$\mathcal{H}_1, \mathcal{H}_2 \to$ hyperplanes

$n_1, n_2 \to$ normal vectors.

I. $l_1, l_2 \to$ 2 suplementary angles: $\measuredangle(v_1,v_2)$, $\measuredangle(-v_1,v_2)$:

$$\cos \measuredangle(v_1,v_2) = \frac{\langle v_1, v_2\rangle}{|v_1|\cdot|v_2|}.$$

II. $\mathcal{H}_1, \mathcal{H}_2 \to$ 2 suplem. angles: $\measuredangle(n_1,n_2)$, $\measuredangle(-n_1,n_2)$:

$$\cos \measuredangle(n_1,n_2) = \frac{\langle n_1, n_2\rangle}{|n_1|\cdot|n_2|}$$

III. $l_1, \mathcal{H}_1 \to$ 2 suplem. angles:

- if $\cos\measuredangle(v_1, n_1) \geq 0 \Rightarrow \measuredangle(v_1,n_1)$ acute.

acute angle between l_1 and $\mathcal{H}_1$:

$$\frac{\pi}{2} - \arccos\left(\frac{\langle v_1, n_1\rangle}{|v_1|\cdot|n_1|}\right).$$

- if $\cos\measuredangle(v_1,n_1) < 0 \Rightarrow$ replace n_1 with normal vector $-n_1$ of $\mathcal{H}_1$.

$\mathbb{E}^3 \to$ dihedral angles.

Π_1, Π_2 define 4 dihedral angles.
(planes)

$\Pi_1 \cap \Pi_2 = \{l\}$ (line).

$\Pi \to$ orthogonal to l.

$l_1 = \Pi \cap \Pi_1$
$l_2 = \Pi \cap \Pi_2$ $>$ $\measuredangle\, l_1, l_2$ don't depend on Π.